

Functions of Several Variables

KEY TERMS

boundary point a point P_0 of R is a boundary point if every δ disk centered around P_0 contains points both inside and outside R

closed set a set S that contains all its boundary points

connected set an open set S that cannot be represented as the union of two or more disjoint, nonempty open subsets

constraint an inequality or equation involving one or more variables that is used in an optimization problem; the constraint enforces a limit on the possible solutions for the problem

contour map a plot of the various level curves of a given function $f(x, y)$

critical point of a function of two variables the point (x_0, y_0) is called a critical point of $f(x, y)$ if one of the two following conditions holds:

1. $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$
2. At least one of $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ do not exist

differentiable a function $f(x, y)$ is differentiable at (x_0, y_0) if $f(x, y)$ can be expressed in the form $f(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + E(x, y)$,

where the error term $E(x, y)$ satisfies $\lim_{(x, y) \rightarrow (x_0, y_0)} \frac{E(x, y)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} = 0$

directional derivative the derivative of a function in the direction of a given unit vector

discriminant the discriminant of the function $f(x, y)$ is given by the formula

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$$

function of two variables a function $z = f(x, y)$ that maps each ordered pair (x, y) in a subset D of $\mathbb{R}^2$ to a unique real number z

generalized chain rule the chain rule extended to functions of more than one independent variable, in which each independent variable may depend on one or more other variables

gradient the gradient of the function $f(x, y)$ is defined to be $\nabla f(x, y) = (\partial f / \partial x)\mathbf{i} + (\partial f / \partial y)\mathbf{j}$, which can be generalized to a function of any number of independent variables

graph of a function of two variables a set of ordered triples (x, y, z) that satisfies the equation $z = f(x, y)$ plotted in three-dimensional Cartesian space

higher-order partial derivatives second-order or higher partial derivatives, regardless of whether they are mixed partial derivatives

interior point a point P_0 of R is a boundary point if there is a δ disk centered around P_0 contained completely in R

intermediate variable given a composition of functions (e.g., $f(x(t), y(t))$), the intermediate variables are the variables that are independent in the outer function but dependent on other variables as well; in the function $f(x(t), y(t))$, the variables x and y are examples of intermediate variables

Lagrange multiplier the constant (or constants) used in the method of Lagrange multipliers; in the case of one constant, it is represented by the variable λ

level curve of a function of two variables the set of points satisfying the equation $f(x, y) = c$ for some real

number c in the range of f

level surface of a function of three variables the set of points satisfying the equation $f(x, y, z) = c$ for some real number c in the range of f

linear approximation given a function $f(x, y)$ and a tangent plane to the function at a point (x_0, y_0) , we can approximate $f(x, y)$ for points near (x_0, y_0) using the tangent plane formula

method of Lagrange multipliers a method of solving an optimization problem subject to one or more constraints

mixed partial derivatives second-order or higher partial derivatives, in which at least two of the differentiations are with respect to different variables

objective function the function that is to be maximized or minimized in an optimization problem

open set a set S that contains none of its boundary points

optimization problem calculation of a maximum or minimum value of a function of several variables, often using Lagrange multipliers

partial derivative a derivative of a function of more than one independent variable in which all the variables but one are held constant

partial differential equation an equation that involves an unknown function of more than one independent variable and one or more of its partial derivatives

region an open, connected, nonempty subset of $\mathbb{R}^2$

saddle point given the function $z = f(x, y)$, the point $(x_0, y_0, f(x_0, y_0))$ is a saddle point if both $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$, but f does not have a local extremum at (x_0, y_0)

surface the graph of a function of two variables, $z = f(x, y)$

tangent plane given a function $f(x, y)$ that is differentiable at a point (x_0, y_0) , the equation of the tangent plane to the surface $z = f(x, y)$ is given by $z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

total differential the total differential of the function $f(x, y)$ at (x_0, y_0) is given by the formula $dz = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy$

tree diagram illustrates and derives formulas for the generalized chain rule, in which each independent variable is accounted for

vertical trace the set of ordered triples (c, y, z) that solves the equation $f(c, y) = z$ for a given constant $x = c$ or the set of ordered triples (x, d, z) that solves the equation $f(x, d) = z$ for a given constant $y = d$

δ ball all points in $\mathbb{R}^3$ lying at a distance of less than δ from (x_0, y_0, z_0)

δ disk an open disk of radius δ centered at point (a, b)

KEY EQUATIONS

- **Vertical trace**
 $f(a, y) = z$ for $x = a$ or $f(x, b) = z$ for $y = b$
- **Level surface of a function of three variables**
 $f(x, y, z) = c$
- **Partial derivative of f with respect to x**

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

- **Partial derivative of f with respect to y**

$$\frac{\partial f}{\partial y} = \lim_{k \rightarrow 0} \frac{f(x, y+k) - f(x, y)}{k}$$

- **Tangent plane**

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

- **Linear approximation**

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

- **Total differential**

$$dz = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy.$$

- **Differentiability (two variables)**

$$f(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + E(x, y),$$

where the error term E satisfies

$$\lim_{(x, y) \rightarrow (x_0, y_0)} \frac{E(x, y)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} = 0.$$

- **Differentiability (three variables)**

$$f(x, y, z) = f(x_0, y_0, z_0) + f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0) + E(x, y, z),$$

where the error term E satisfies

$$\lim_{(x, y, z) \rightarrow (x_0, y_0, z_0)} \frac{E(x, y, z)}{\sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2}} = 0.$$

- **Chain rule, one independent variable**

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

- **Chain rule, two independent variables**

$$\frac{dz}{du} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{dz}{dv} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

- **Generalized chain rule**

$$\frac{\partial w}{\partial t_j} = \frac{\partial w}{\partial x_1} \frac{\partial x_1}{\partial t_j} + \frac{\partial w}{\partial x_2} \frac{\partial x_2}{\partial t_j} + \dots + \frac{\partial w}{\partial x_m} \frac{\partial x_m}{\partial t_j}$$

- **directional derivative (two dimensions)**

$$D_{\mathbf{u}}f(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h \cos \theta, b + h \sin \theta) - f(a, b)}{h}$$

or

$$D_{\mathbf{u}}f(x, y) = f_x(x, y)\cos \theta + f_y(x, y)\sin \theta$$

- **gradient (two dimensions)**

$$\nabla f(x, y) = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}$$

- **gradient (three dimensions)**

$$\nabla f(x, y, z) = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}$$

- **directional derivative (three dimensions)**

$$D_{\mathbf{u}}f(x, y, z) = \nabla f(x, y, z) \cdot \mathbf{u} \\ = f_x(x, y, z)\cos \alpha + f_y(x, y, z)\cos \beta + f_z(x, y, z)\cos \gamma$$

- **Discriminant**

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$$

- **Method of Lagrange multipliers, one constraint**

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0)$$

$$g(x_0, y_0) = 0$$

- **Method of Lagrange multipliers, two constraints**

$$\nabla f(x_0, y_0, z_0) = \lambda_1 \nabla g(x_0, y_0, z_0) + \lambda_2 \nabla h(x_0, y_0, z_0)$$

$$g(x_0, y_0, z_0) = 0$$

$$h(x_0, y_0, z_0) = 0$$

KEY CONCEPTS

4.1 Functions of Several Variables

- The graph of a function of two variables is a surface in $\mathbb{R}^3$ and can be studied using level curves and vertical traces.
- A set of level curves is called a contour map.

4.2 Limits and Continuity

- To study limits and continuity for functions of two variables, we use a δ disk centered around a given point.
- A function of several variables has a limit if for any point in a δ ball centered at a point P , the value of the function at that point is arbitrarily close to a fixed value (the limit value).
- The limit laws established for a function of one variable have natural extensions to functions of more than one variable.
- A function of two variables is continuous at a point if the limit exists at that point, the function exists at that point, and the limit and function are equal at that point.

4.3 Partial Derivatives

- A partial derivative is a derivative involving a function of more than one independent variable.
- To calculate a partial derivative with respect to a given variable, treat all the other variables as constants and use the usual differentiation rules.
- Higher-order partial derivatives can be calculated in the same way as higher-order derivatives.

4.4 Tangent Planes and Linear Approximations

- The analog of a tangent line to a curve is a tangent plane to a surface for functions of two variables.
- Tangent planes can be used to approximate values of functions near known values.
- A function is differentiable at a point if it is "smooth" at that point (i.e., no corners or discontinuities exist at that point).
- The total differential can be used to approximate the change in a function $z = f(x_0, y_0)$ at the point (x_0, y_0) for given values of Δx and Δy .

4.5 The Chain Rule

- The chain rule for functions of more than one variable involves the partial derivatives with respect to all the independent variables.
- Tree diagrams are useful for deriving formulas for the chain rule for functions of more than one variable, where each independent variable also depends on other variables.

4.6 Directional Derivatives and the Gradient

- A directional derivative represents a rate of change of a function in any given direction.
- The gradient can be used in a formula to calculate the directional derivative.
- The gradient indicates the direction of greatest change of a function of more than one variable.

4.7 Maxima/Minima Problems

- A critical point of the function $f(x, y)$ is any point (x_0, y_0) where either $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$, or at least one of $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ do not exist.
- A saddle point is a point (x_0, y_0) where $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$, but (x_0, y_0) is neither a maximum nor a minimum at that point.
- To find extrema of functions of two variables, first find the critical points, then calculate the discriminant and apply the second derivative test.

4.8 Lagrange Multipliers

- An objective function combined with one or more constraints is an example of an optimization problem.
- To solve optimization problems, we apply the method of Lagrange multipliers using a four-step problem-solving strategy.

Multiple Integrals

KEY TERMS

double integral of the function $f(x, y)$ over the region R in the xy -plane is defined as the limit of a double Riemann

$$\text{sum, } \iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A.$$

double Riemann sum of the function $f(x, y)$ over a rectangular region R is $\sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$ where R is divided into smaller subrectangles R_{ij} and (x_{ij}^*, y_{ij}^*) is an arbitrary point in R_{ij}

Fubini's theorem if $f(x, y)$ is a function of two variables that is continuous over a rectangular region

$R = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, c \leq y \leq d\}$, then the double integral of f over the region equals an iterated integral,

$$\iint_R f(x, y) dy dx = \int_a^b \int_c^d f(x, y) dx dy = \int_c^d \int_a^b f(x, y) dx dy$$

improper double integral a double integral over an unbounded region or of an unbounded function

iterated integral for a function $f(x, y)$ over the region R is

$$\text{a. } \int_a^b \int_c^d f(x, y) dx dy = \int_a^b \left[\int_c^d f(x, y) dy \right] dx,$$

$$\text{b. } \int_c^d \int_a^b f(x, y) dx dy = \int_c^d \left[\int_a^b f(x, y) dx \right] dy,$$

where $a, b, c,$ and d are any real numbers and $R = [a, b] \times [c, d]$

Jacobian the Jacobian $J(u, v)$ in two variables is a 2×2 determinant:

$$J(u, v) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix};$$

the Jacobian $J(u, v, w)$ in three variables is a 3×3 determinant:

$$J(u, v, w) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \\ \frac{\partial x}{\partial w} & \frac{\partial y}{\partial w} & \frac{\partial z}{\partial w} \end{vmatrix}$$

one-to-one transformation a transformation $T : G \rightarrow R$ defined as $T(u, v) = (x, y)$ is said to be one-to-one if no two points map to the same image point

planar transformation a function T that transforms a region G in one plane into a region R in another plane by a change of variables

polar rectangle the region enclosed between the circles $r = a$ and $r = b$ and the angles $\theta = \alpha$ and $\theta = \beta$; it is described as $R = \{(r, \theta) \mid a \leq r \leq b, \alpha \leq \theta \leq \beta\}$

radius of gyration the distance from an object's center of mass to its axis of rotation

transformation a function that transforms a region G in one plane into a region R in another plane by a change of variables

triple integral the triple integral of a continuous function $f(x, y, z)$ over a rectangular solid box B is the limit of a Riemann sum for a function of three variables, if this limit exists

triple integral in cylindrical coordinates the limit of a triple Riemann sum, provided the following limit exists:

$$\lim_{l, m, n \rightarrow \infty} \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(r_{ijk}^*, \theta_{ijk}^*, z_{ijk}^*) r_{ijk}^* \Delta r \Delta \theta \Delta z$$

triple integral in spherical coordinates the limit of a triple Riemann sum, provided the following limit exists:

$$\lim_{l, m, n \rightarrow \infty} \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(\rho_{ijk}^*, \theta_{ijk}^*, \varphi_{ijk}^*) (\rho_{ijk}^*)^2 \sin \varphi \Delta \rho \Delta \theta \Delta \varphi$$

Type I a region D in the xy -plane is Type I if it lies between two vertical lines and the graphs of two continuous functions $g_1(x)$ and $g_2(x)$

Type II a region D in the xy -plane is Type II if it lies between two horizontal lines and the graphs of two continuous functions $h_1(y)$ and $h_2(y)$

KEY EQUATIONS

- Double integral**

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

- Iterated integral**

$$\int_a^b \int_c^d f(x, y) dx dy = \int_a^b \left[\int_c^d f(x, y) dy \right] dx$$

or

$$\int_c^d \int_a^b f(x, y) dx dy = \int_c^d \left[\int_a^b f(x, y) dx \right] dy$$

- Average value of a function of two variables**

$$f_{\text{ave}} = \frac{1}{\text{Area } R} \iint_R f(x, y) dx dy$$

- Iterated integral over a Type I region**

$$\iint_D f(x, y) dA = \iint_D f(x, y) dy dx = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x, y) dy \right] dx$$

- Iterated integral over a Type II region**

$$\iint_D f(x, y) dA = \iint_D f(x, y) dx dy = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} f(x, y) dx \right] dy$$

- Double integral over a polar rectangular region R**

$$\iint_R f(r, \theta) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(r_{ij}^*, \theta_{ij}^*) \Delta A = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(r_{ij}^*, \theta_{ij}^*) r_{ij}^* \Delta r \Delta \theta$$

- Double integral over a general polar region**

$$\iint_D f(r, \theta) r dr d\theta = \int_{\theta=\alpha}^{\theta=\beta} \int_{r=h_1(\theta)}^{r=h_2(\theta)} f(r, \theta) r dr d\theta$$

- **Triple integral**

$$\lim_{l, m, n \rightarrow \infty} \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta x \Delta y \Delta z = \iiint_B f(x, y, z) dV$$

- **Triple integral in cylindrical coordinates**

$$\iiint_B g(x, y, z) dV = \iiint_B g(r \cos \theta, r \sin \theta, z) r dr d\theta dz = \iiint_B f(r, \theta, z) r dr d\theta dz$$

- **Triple integral in spherical coordinates**

$$\iiint_B f(\rho, \theta, \varphi) \rho^2 \sin \varphi d\rho d\varphi d\theta = \int_{\varphi=\gamma}^{\varphi=\psi} \int_{\theta=\alpha}^{\theta=\beta} \int_{\rho=a}^{\rho=b} f(\rho, \theta, \varphi) \rho^2 \sin \varphi d\rho d\varphi d\theta$$

- **Mass of a lamina**

$$m = \lim_{k, l \rightarrow \infty} \sum_{i=1}^k \sum_{j=1}^l m_{ij} = \lim_{k, l \rightarrow \infty} \sum_{i=1}^k \sum_{j=1}^l \rho(x_{ij}^*, y_{ij}^*) \Delta A = \iint_R \rho(x, y) dA$$

- **Moment about the x-axis**

$$M_x = \lim_{k, l \rightarrow \infty} \sum_{i=1}^k \sum_{j=1}^l (y_{ij}^*) m_{ij} = \lim_{k, l \rightarrow \infty} \sum_{i=1}^k \sum_{j=1}^l (y_{ij}^*) \rho(x_{ij}^*, y_{ij}^*) \Delta A = \iint_R y \rho(x, y) dA$$

- **Moment about the y-axis**

$$M_y = \lim_{k, l \rightarrow \infty} \sum_{i=1}^k \sum_{j=1}^l (x_{ij}^*) m_{ij} = \lim_{k, l \rightarrow \infty} \sum_{i=1}^k \sum_{j=1}^l (x_{ij}^*) \rho(x_{ij}^*, y_{ij}^*) \Delta A = \iint_R x \rho(x, y) dA$$

- **Center of mass of a lamina**

$$\bar{x} = \frac{M_y}{m} = \frac{\iint_R x \rho(x, y) dA}{\iint_R \rho(x, y) dA} \quad \text{and} \quad \bar{y} = \frac{M_x}{m} = \frac{\iint_R y \rho(x, y) dA}{\iint_R \rho(x, y) dA}$$

KEY CONCEPTS

5.1 Double Integrals over Rectangular Regions

- We can use a double Riemann sum to approximate the volume of a solid bounded above by a function of two variables over a rectangular region. By taking the limit, this becomes a double integral representing the volume of the solid.
- Properties of double integral are useful to simplify computation and find bounds on their values.
- We can use Fubini's theorem to write and evaluate a double integral as an iterated integral.
- Double integrals are used to calculate the area of a region, the volume under a surface, and the average value of a function of two variables over a rectangular region.

5.2 Double Integrals over General Regions

- A general bounded region D on the plane is a region that can be enclosed inside a rectangular region. We can use this idea to define a double integral over a general bounded region.
- To evaluate an iterated integral of a function over a general nonrectangular region, we sketch the region and express it as a Type I or as a Type II region or as a union of several Type I or Type II regions that overlap only on their boundaries.

- We can use double integrals to find volumes, areas, and average values of a function over general regions, similarly to calculations over rectangular regions.
- We can use Fubini's theorem for improper integrals to evaluate some types of improper integrals.

5.3 Double Integrals in Polar Coordinates

- To apply a double integral to a situation with circular symmetry, it is often convenient to use a double integral in polar coordinates. We can apply these double integrals over a polar rectangular region or a general polar region, using an iterated integral similar to those used with rectangular double integrals.
- The area dA in polar coordinates becomes $r dr d\theta$.
- Use $x = r \cos \theta$, $y = r \sin \theta$, and $dA = r dr d\theta$ to convert an integral in rectangular coordinates to an integral in polar coordinates.
- Use $r^2 = x^2 + y^2$ and $\theta = \tan^{-1}\left(\frac{y}{x}\right)$ to convert an integral in polar coordinates to an integral in rectangular coordinates, if needed.
- To find the volume in polar coordinates bounded above by a surface $z = f(r, \theta)$ over a region on the xy -plane, use a double integral in polar coordinates.

5.4 Triple Integrals

- To compute a triple integral we use Fubini's theorem, which states that if $f(x, y, z)$ is continuous on a rectangular box $B = [a, b] \times [c, d] \times [e, f]$, then

$$\iiint_B f(x, y, z) dV = \int_e^f \int_c^d \int_a^b f(x, y, z) dx dy dz$$

and is also equal to any of the other five possible orderings for the iterated triple integral.

- To compute the volume of a general solid bounded region E we use the triple integral

$$V(E) = \iiint_E 1 dV.$$

- Interchanging the order of the iterated integrals does not change the answer. As a matter of fact, interchanging the order of integration can help simplify the computation.
- To compute the average value of a function over a general three-dimensional region, we use

$$f_{\text{ave}} = \frac{1}{V(E)} \iiint_E f(x, y, z) dV.$$

5.5 Triple Integrals in Cylindrical and Spherical Coordinates

- To evaluate a triple integral in cylindrical coordinates, use the iterated integral

$$\int_{\theta=\alpha}^{\theta=\beta} \int_{r=g_1(\theta)}^{r=g_2(\theta)} \int_{z=u_1(r, \theta)}^{z=u_2(r, \theta)} f(r, \theta, z) r dz dr d\theta.$$

- To evaluate a triple integral in spherical coordinates, use the iterated integral

$$\int_{\theta=\alpha}^{\theta=\beta} \int_{\rho=g_1(\theta)}^{\rho=g_2(\theta)} \int_{\varphi=u_1(r, \theta)}^{\varphi=u_2(r, \theta)} f(\rho, \theta, \varphi) \rho^2 \sin \varphi d\varphi d\rho d\theta.$$

5.6 Calculating Centers of Mass and Moments of Inertia

Finding the mass, center of mass, moments, and moments of inertia in double integrals:

- For a lamina R with a density function $\rho(x, y)$ at any point (x, y) in the plane, the mass is $m = \iint_R \rho(x, y) dA$.
- The moments about the x -axis and y -axis are

$$M_x = \iint_R y\rho(x, y) dA \text{ and } M_y = \iint_R x\rho(x, y) dA.$$

- The center of mass is given by $\bar{x} = \frac{M_y}{m}$, $\bar{y} = \frac{M_x}{m}$.
- The center of mass becomes the centroid of the plane when the density is constant.
- The moments of inertia about the x -axis, y -axis, and the origin are

$$I_x = \iint_R y^2 \rho(x, y) dA, \quad I_y = \iint_R x^2 \rho(x, y) dA, \quad \text{and } I_0 = I_x + I_y = \iint_R (x^2 + y^2) \rho(x, y) dA.$$

Finding the mass, center of mass, moments, and moments of inertia in triple integrals:

- For a solid object Q with a density function $\rho(x, y, z)$ at any point (x, y, z) in space, the mass is $m = \iiint_Q \rho(x, y, z) dV$.

- The moments about the xy -plane, the xz -plane, and the yz -plane are

$$M_{xy} = \iiint_Q z\rho(x, y, z) dV, \quad M_{xz} = \iiint_Q y\rho(x, y, z) dV, \quad M_{yz} = \iiint_Q x\rho(x, y, z) dV.$$

- The center of mass is given by $\bar{x} = \frac{M_{yz}}{m}$, $\bar{y} = \frac{M_{xz}}{m}$, $\bar{z} = \frac{M_{xy}}{m}$.
- The center of mass becomes the centroid of the solid when the density is constant.
- The moments of inertia about the yz -plane, the xz -plane, and the xy -plane are

$$I_x = \iiint_Q (y^2 + z^2) \rho(x, y, z) dV, \quad I_y = \iiint_Q (x^2 + z^2) \rho(x, y, z) dV, \\ I_z = \iiint_Q (x^2 + y^2) \rho(x, y, z) dV.$$

5.7 Change of Variables in Multiple Integrals

- A transformation T is a function that transforms a region G in one plane (space) into a region R in another plane (space) by a change of variables.
- A transformation $T : G \rightarrow R$ defined as $T(u, v) = (x, y)$ (or $T(u, v, w) = (x, y, z)$) is said to be a one-to-one transformation if no two points map to the same image point.
- If f is continuous on R , then $\iint_R f(x, y) dA = \iint_S f(g(u, v), h(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$.
- If F is continuous on R , then

$$\begin{aligned} \iiint_R F(x, y, z) dV &= \iiint_G F(g(u, v, w), h(u, v, w), k(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw \\ &= \iiint_G H(u, v, w) |J(u, v, w)| du dv dw. \end{aligned}$$

Vector Calculus

KEY TERMS

circulation the tendency of a fluid to move in the direction of curve C . If C is a closed curve, then the circulation of $\mathbf{F}$ along C is line integral $\int_C \mathbf{F} \cdot \mathbf{T} ds$, which we also denote $\oint_C \mathbf{F} \cdot \mathbf{T} ds$

closed curve a curve for which there exists a parameterization $\mathbf{r}(t)$, $a \leq t \leq b$, such that $\mathbf{r}(a) = \mathbf{r}(b)$, and the curve is traversed exactly once

closed curve a curve that begins and ends at the same point

connected region a region in which any two points can be connected by a path with a trace contained entirely inside the region

conservative field a vector field for which there exists a scalar function f such that $\nabla f = \mathbf{F}$

curl

the curl of vector field $\mathbf{F} = \langle P, Q, R \rangle$, denoted $\nabla \times \mathbf{F}$, is the “determinant” of the matrix $\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$ and is

given by the expression $(R_y - Q_z)\mathbf{i} + (P_z - R_x)\mathbf{j} + (Q_x - P_y)\mathbf{k}$; it measures the tendency of particles at a point to rotate about the axis that points in the direction of the curl at the point

divergence the divergence of a vector field $\mathbf{F} = \langle P, Q, R \rangle$, denoted $\nabla \cdot \mathbf{F}$, is $P_x + Q_y + R_z$; it measures the “outflowing-ness” of a vector field

divergence theorem a theorem used to transform a difficult flux integral into an easier triple integral and vice versa

flux the rate of a fluid flowing across a curve in a vector field; the flux of vector field $\mathbf{F}$ across plane curve C is line integral $\int_C \mathbf{F} \cdot \frac{\mathbf{n}(t)}{\|\mathbf{n}(t)\|} ds$

flux integral another name for a surface integral of a vector field; the preferred term in physics and engineering

Fundamental Theorem for Line Integrals the value of line integral $\int_C \nabla f \cdot d\mathbf{r}$ depends only on the value of f at the endpoints of C : $\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$

Gauss' law if S is a piecewise, smooth closed surface in a vacuum and Q is the total stationary charge inside of S , then the flux of electrostatic field $\mathbf{E}$ across S is Q/ϵ_0

gradient field a vector field $\mathbf{F}$ for which there exists a scalar function f such that $\nabla f = \mathbf{F}$; in other words, a vector field that is the gradient of a function; such vector fields are also called *conservative*

Green's theorem relates the integral over a connected region to an integral over the boundary of the region

grid curves curves on a surface that are parallel to grid lines in a coordinate plane

heat flow a vector field proportional to the negative temperature gradient in an object

independence of path a vector field $\mathbf{F}$ has path independence if $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ for any curves C_1 and C_2 in the domain of $\mathbf{F}$ with the same initial points and terminal points

inverse-square law the electrostatic force at a given point is inversely proportional to the square of the distance from the source of the charge

line integral the integral of a function along a curve in a plane or in space

mass flux the rate of mass flow of a fluid per unit area, measured in mass per unit time per unit area

orientation of a curve the orientation of a curve C is a specified direction of C

orientation of a surface if a surface has an “inner” side and an “outer” side, then an orientation is a choice of the inner or the outer side; the surface could also have “upward” and “downward” orientations

parameter domain (parameter space) the region of the uv plane over which the parameters u and v vary for parameterization $\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$

parameterized surface (parametric surface) a surface given by a description of the form $\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$, where the parameters u and v vary over a parameter domain in the uv -plane

piecewise smooth curve an oriented curve that is not smooth, but can be written as the union of finitely many smooth curves

potential function a scalar function f such that $\nabla f = \mathbf{F}$

radial field a vector field in which all vectors either point directly toward or directly away from the origin; the magnitude of any vector depends only on its distance from the origin

regular parameterization parameterization $\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$ such that $\mathbf{r}_u \times \mathbf{r}_v$ is not zero for point (u, v) in the parameter domain

rotational field a vector field in which the vector at point (x, y) is tangent to a circle with radius $r = \sqrt{x^2 + y^2}$; in a rotational field, all vectors flow either clockwise or counterclockwise, and the magnitude of a vector depends only on its distance from the origin

scalar line integral the scalar line integral of a function f along a curve C with respect to arc length is the integral

$\int_C f ds$, it is the integral of a scalar function f along a curve in a plane or in space; such an integral is defined in terms of a Riemann sum, as is a single-variable integral

simple curve a curve that does not cross itself

simply connected region a region that is connected and has the property that any closed curve that lies entirely inside the region encompasses points that are entirely inside the region

Stokes' theorem relates the flux integral over a surface S to a line integral around the boundary C of the surface S

stream function if $\mathbf{F} = \langle P, Q \rangle$ is a source-free vector field, then stream function g is a function such that $P = g_y$ and $Q = -g_x$

surface area the area of surface S given by the surface integral $\iint_S dS$

surface independent flux integrals of curl vector fields are surface independent if their evaluation does not depend on the surface but only on the boundary of the surface

surface integral an integral of a function over a surface

surface integral of a scalar-valued function a surface integral in which the integrand is a scalar function

surface integral of a vector field a surface integral in which the integrand is a vector field

unit vector field a vector field in which the magnitude of every vector is 1

vector field measured in $\mathbb{R}^2$, an assignment of a vector $\mathbf{F}(x, y)$ to each point (x, y) of a subset D of $\mathbb{R}^2$; in $\mathbb{R}^3$, an assignment of a vector $\mathbf{F}(x, y, z)$ to each point (x, y, z) of a subset D of $\mathbb{R}^3$

vector line integral the vector line integral of vector field $\mathbf{F}$ along curve C is the integral of the dot product of $\mathbf{F}$ with unit tangent vector $\mathbf{T}$ of C with respect to arc length, $\int_C \mathbf{F} \cdot \mathbf{T} ds$; such an integral is defined in terms of a Riemann

sum, similar to a single-variable integral

KEY EQUATIONS

- **Vector field in $\mathbb{R}^2$**

$$\mathbf{F}(x, y) = \langle P(x, y), Q(x, y) \rangle$$

or

$$\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$$

- **Vector field in $\mathbb{R}^3$**

$$\mathbf{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle$$

or

$$\mathbf{F}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k}$$

- **Calculating a scalar line integral**

$$\int_C f(x, y, z) ds = \int_a^b f(\mathbf{r}(t)) \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} dt$$

- **Calculating a vector line integral**

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_C \mathbf{F} \cdot \mathbf{T} ds = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

or

$$\int_C P dx + Q dy + R dz = \int_a^b \left(P(\mathbf{r}(t)) \frac{dx}{dt} + Q(\mathbf{r}(t)) \frac{dy}{dt} + R(\mathbf{r}(t)) \frac{dz}{dt} \right) dt$$

- **Calculating flux**

$$\int_C \mathbf{F} \cdot \frac{\mathbf{n}(t)}{\|\mathbf{n}(t)\|} ds = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{n}(t) dt$$

- **Fundamental Theorem for Line Integrals**

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

- **Circulation of a conservative field over curve C that encloses a simply connected region**

$$\oint_C \nabla f \cdot d\mathbf{r} = 0$$

- **Green's theorem, circulation form**

$$\oint_C P dx + Q dy = \iint_D Q_x - P_y dA, \quad \text{where } C \text{ is the boundary of } D$$

- **Green's theorem, flux form**

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D Q_x - P_y dA, \quad \text{where } C \text{ is the boundary of } D$$

- **Green's theorem, extended version**

$$\oint_{\partial D} \mathbf{F} \cdot d\mathbf{r} = \iint_D Q_x - P_y dA$$

- **Curl**

$$\nabla \times \mathbf{F} = (R_y - Q_z)\mathbf{i} + (P_z - R_x)\mathbf{j} + (Q_x - P_y)\mathbf{k}$$

- **Divergence**

$$\nabla \cdot \mathbf{F} = P_x + Q_y + R_z$$

- **Divergence of curl is zero**

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0$$

- **Curl of a gradient is the zero vector**

$$\nabla \times (\nabla f) = 0$$

- **Scalar surface integral**

$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{t}_u \times \mathbf{t}_v\| dA$$

- **Flux integral**

$$\iint_S \mathbf{F} \cdot \mathbf{N} dS = \iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{t}_u \times \mathbf{t}_v) dA$$

- **Stokes' theorem**

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$$

- **Divergence theorem**

$$\iiint_E \text{div } \mathbf{F} dV = \iint_S \mathbf{F} \cdot d\mathbf{S}$$

KEY CONCEPTS

6.1 Vector Fields

- A vector field assigns a vector $\mathbf{F}(x, y)$ to each point (x, y) in a subset D of $\mathbb{R}^2$ or $\mathbf{F}(x, y, z)$ to each point (x, y, z) in a subset D of $\mathbb{R}^3$.
- Vector fields can describe the distribution of vector quantities such as forces or velocities over a region of the plane or of space. They are in common use in such areas as physics, engineering, meteorology, oceanography.
- We can sketch a vector field by examining its defining equation to determine relative magnitudes in various locations and then drawing enough vectors to determine a pattern.
- A vector field $\mathbf{F}$ is called conservative if there exists a scalar function f such that $\nabla f = \mathbf{F}$.

6.2 Line Integrals

- Line integrals generalize the notion of a single-variable integral to higher dimensions. The domain of integration in a single-variable integral is a line segment along the x -axis, but the domain of integration in a line integral is a curve in a plane or in space.
- If C is a curve, then the length of C is $\int_C ds$.
- There are two kinds of line integral: scalar line integrals and vector line integrals. Scalar line integrals can be used to calculate the mass of a wire; vector line integrals can be used to calculate the work done on a particle traveling through a field.
- Scalar line integrals can be calculated using **Equation 6.8**; vector line integrals can be calculated using **Equation 6.9**.
- Two key concepts expressed in terms of line integrals are flux and circulation. Flux measures the rate that a field crosses a given line; circulation measures the tendency of a field to move in the same direction as a given closed curve.

6.3 Conservative Vector Fields

- The theorems in this section require curves that are closed, simple, or both, and regions that are connected or simply connected.
- The line integral of a conservative vector field can be calculated using the Fundamental Theorem for Line Integrals. This theorem is a generalization of the Fundamental Theorem of Calculus in higher dimensions. Using this theorem usually makes the calculation of the line integral easier.
- Conservative fields are independent of path. The line integral of a conservative field depends only on the value of

the potential function at the endpoints of the domain curve.

- Given vector field $\mathbf{F}$, we can test whether $\mathbf{F}$ is conservative by using the cross-partial property. If $\mathbf{F}$ has the cross-partial property and the domain is simply connected, then $\mathbf{F}$ is conservative (and thus has a potential function). If $\mathbf{F}$ is conservative, we can find a potential function by using the Problem-Solving Strategy.
- The circulation of a conservative vector field on a simply connected domain over a closed curve is zero.

6.4 Green's Theorem

- Green's theorem relates the integral over a connected region to an integral over the boundary of the region. Green's theorem is a version of the Fundamental Theorem of Calculus in one higher dimension.
- Green's Theorem comes in two forms: a circulation form and a flux form. In the circulation form, the integrand is $\mathbf{F} \cdot \mathbf{T}$. In the flux form, the integrand is $\mathbf{F} \cdot \mathbf{N}$.
- Green's theorem can be used to transform a difficult line integral into an easier double integral, or to transform a difficult double integral into an easier line integral.
- A vector field is source free if it has a stream function. The flux of a source-free vector field across a closed curve is zero, just as the circulation of a conservative vector field across a closed curve is zero.

6.5 Divergence and Curl

- The divergence of a vector field is a scalar function. Divergence measures the “outflowing-ness” of a vector field. If $\mathbf{v}$ is the velocity field of a fluid, then the divergence of $\mathbf{v}$ at a point is the outflow of the fluid less the inflow at the point.
- The curl of a vector field is a vector field. The curl of a vector field at point P measures the tendency of particles at P to rotate about the axis that points in the direction of the curl at P .
- A vector field with a simply connected domain is conservative if and only if its curl is zero.

6.6 Surface Integrals

- Surfaces can be parameterized, just as curves can be parameterized. In general, surfaces must be parameterized with two parameters.
- Surfaces can sometimes be oriented, just as curves can be oriented. Some surfaces, such as a Möbius strip, cannot be oriented.
- A surface integral is like a line integral in one higher dimension. The domain of integration of a surface integral is a surface in a plane or space, rather than a curve in a plane or space.
- The integrand of a surface integral can be a scalar function or a vector field. To calculate a surface integral with an integrand that is a function, use **Equation 6.19**. To calculate a surface integral with an integrand that is a vector field, use **Equation 6.20**.
- If S is a surface, then the area of S is $\iint_S dS$.

6.7 Stokes' Theorem

- Stokes' theorem relates a flux integral over a surface to a line integral around the boundary of the surface. Stokes' theorem is a higher dimensional version of Green's theorem, and therefore is another version of the Fundamental Theorem of Calculus in higher dimensions.
- Stokes' theorem can be used to transform a difficult surface integral into an easier line integral, or a difficult line integral into an easier surface integral.
- Through Stokes' theorem, line integrals can be evaluated using the simplest surface with boundary C .
- Faraday's law relates the curl of an electric field to the rate of change of the corresponding magnetic field. Stokes' theorem can be used to derive Faraday's law.

6.8 The Divergence Theorem

- The divergence theorem relates a surface integral across closed surface S to a triple integral over the solid enclosed by S . The divergence theorem is a higher dimensional version of the flux form of Green's theorem, and is therefore a higher dimensional version of the Fundamental Theorem of Calculus.
- The divergence theorem can be used to transform a difficult flux integral into an easier triple integral and vice versa.
- The divergence theorem can be used to derive Gauss' law, a fundamental law in electrostatics.